Homework Assignment 5

Math 311, Fall 2026

Rutgers University

Exercise 2.7.1 Let (a_n) be a non-negative decreasing sequnce such that $a_n \to 0$. Define the sequence of partial sums as

$$s_n = a_1 - a_2 + a_3 - \cdots \pm a_n.$$

(a) Let $\epsilon > 0$. Since $a_n \to 0$, there exists $N \in \mathbb{N}$ such that

$$a_n < \epsilon.$$

for all $n \geq N$. Let $m > n \geq N$. Then

$$|s_m - s_n| = |a_{n+1} - a_{n+2} + \cdots \pm a_m|.$$

Group consecutive terms

$$(a_{n+1} - a_{n+2}) + (a_{n+3} - a_{n+4}) + \ldots$$

Since a_n is decreasing and non-negative, each grouping must also be non-negative, and so the sum is non-negative. Moreover, shifting the grouping to the right by one term

$$a_{n+1} - (a_{n+2} - a_{n+3}) - (a_{n+4} - a_{n+5}) + \cdots \geq 0.$$

Thus

$$0 \leq a_{n+1} - a_{n+2} + \cdots \pm a_m \leq a_{n+1},$$

so

$$|a_{n+1} - a_{n+2} + \cdots \pm a_m| \leq a_{n+1}.$$

Therefore,

$$|s_m - s_n| \leq a_{n+1} < \epsilon.$$

Hence s_n is a Cauchy sequence, so by Cauchy Criterion, s_n converges.

(b) For every $n \in \mathbb{N}$,

$$s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0.$$

Hence even partial sums are increasing. Similarly,

$$s_{2n+3} - s_{2n+1} = a_{2n+3} - a_{2n+2} \leq 0.$$

That is, odd partial sums are decreasing. Finally,

$$s_{2n+1} - s_{2n} = a_{2n+1} \geq 0.$$

Next, we construct our intervals as

$$I_n = [s_{2n}, s_{2n+1}]$$

Then

$$I_{n+1} \subseteq I_n$$

By the Nested Interval Property, there exists $s \in \mathbb{R}$ such that

$$s \in \bigcap_{n=1}^{\infty} I_n.$$

Thus,

$$s_{2n} \leq s \leq s_{2n+1}.$$

Observe that

$$s_{2n+1} - s_{2n} = a_{2n+1}.$$

Because $a_n \to 0$,

$$a_{2n+1} \to 0.$$

Since $s \in I_n$,

$$0 \leq s - s_{2n} \leq a_{2n+1}$$

and

$$0 \leq s_{2n+1} - s \leq a_{2n+1}.$$

By the Squeeze Theorem,

$$s_{2n} \to s \quad \text{and} \quad s_{2n+1} \to s.$$

Because both the even and odd subsequences of (s_n) converge to the same limit s, the entire sequence (s_n) converges to s. Therefore,

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n \text{ converges.}$$

(c) First, observe that

$$s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0.$$

Thus, (s_{2n}) is increasing. Similarly,

$$s_{2n+3} - s_{2n+1} = -a_{2n+2} + a_{2n+3} \leq 0.$$

Thus, (s_{2n+1}) is decreasing. Moreover,

$$s_{2n+1} - s_{2n} = a_{2n+1} \geq 0,$$

so $s_{2n} \leq s_{2n+1}$ for every $n \in \mathbb{N}$. Since (s_{2n}) is increasing and bounded above by s_1, the Monotone Convergence Theorem implies that

$$s_{2n} \to L$$

for some $L \in \mathbb{R}$. Similarly, (s_{2n+1}) is decreasing and bounded below by s_2, so

$$s_{2n+1} \to M$$

for some $M \in \mathbb{R}$. Since

$$s_{2n+1} - s_{2n} = a_{2n+1} \to 0,$$

the Algebraic Limit Theorem gives

$$M - L = 0.$$

Hence, $L = M$. Therefore, both subsequences converge to the same limit. It follows that the entire sequence (s_n) converges to L. Thus,

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n \text{ converges.}$$

Exercise 2.7.2

(a) For every $n \in \mathbb{N}$,

$$\sum_{n=1}^{\infty} \frac{1}{2^n + n} \leq \sum_{n=1}^{\infty} \frac{1}{2^n}$$

Recall that

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \to 1$$

Thus, by comparison test, $\sum_{n=1}^{\infty} \frac{1}{2^n+n}$ converges.

(b) Since $|\sin n| \leq 1$ for every $n \in \mathbb{N}$, it follows that

$$\sum_{n=1}^{\infty} \frac{|\sin n|}{n^2} \leq \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Notice that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent p-series. Hence, by comparison test,

$$\sum_{n=1}^{\infty} \frac{|\sin n|}{n^2}$$

converges. Moreovr, by Absolute Convergence Test,

$$\sum_{n=1}^{\infty} \frac{\sin n}{n^2} \text{ converges.}$$

(c) We can rewrite the series as

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1+n}{2n}$$

Then

$$|a_n| = \frac{1+n}{2n} = \frac{1}{2} + \frac{1}{2n}.$$

So

$$|a_n| \to \frac{1}{2} \neq 0.$$

Hence $a_n \not\to 0$. By the contrapositive of theorem 2.7.3 , we conclude that the series

$$\sum_{n=1}^{\infty}\frac{1+n}{2n} \text{ diverges.}$$

(d) We can group into consecutive terms of 3

$$(1+\frac{1}{2}-\frac{1}{3})+(\frac{1}{4}+\frac{1}{5}-\frac{1}{6})+\dots$$

Then we can define the sequence as

$$a_n = \frac{1}{3n-2}+\frac{1}{3n-1}-\frac{1}{3n}$$

Notice that

$$\frac{1}{3n-1} > \frac{1}{3n}.$$

So

$$a_n \geq \frac{1}{3n-2} \geq \frac{1}{3n}.$$

But

$$\sum_{n=1}^{\infty}\frac{1}{3n} = \frac{1}{3}\sum_{n=1}^{\infty}\frac{1}{n}$$

diverges. So by the comparison test,

$$\sum_{n=1}^{\infty}a_n \quad \text{diverges.}$$

(e) We can group consecutive terms into pairs

$$(1-\frac{1}{2^2})+(\frac{1}{3}-\frac{1}{4^2})+\dots$$

Then we can define the sequence as

$$a_n = \frac{1}{2n-1}-\frac{1}{4n^2}$$

Notice that

$$a_n \geq \frac{1}{2n} - \frac{1}{4n^2} \geq \frac{1}{2n} - \frac{1}{4n} = \frac{1}{4n}.$$

But

$$\sum_{n=1}^{\infty} \frac{1}{4n} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges. So by the comparison test,

$$\sum_{n=1}^{\infty} a_n \qquad \text{diverges.}$$

Exercise 2.7.4

(a) This is possible, consider

$$x_n = y_n = n^{-\frac{3}{4}}.$$

Then both series are p-series with p-value less than 1, so both series diverge. However

$$x_n y_n = n^{-\frac{3}{2}}.$$

Thus, we have a p-series with a p-value greater than one, so the series

$$\sum x_n y_n \quad \text{converges.}$$

(b) This is possible, consider

$$x_n = \frac{(-1)^n}{n} \qquad \text{and} \qquad y_n = (-1)^n.$$

Then $\sum x_n$ converges by the alternating series test, and $|y_n| \leq 1$ so it is bounded. But

$$\sum x_n y_n = \sum \frac{1}{n}.$$

This is the harmonic series, which we know diverges.

(c) This is impossible, suppose

$$\sum x_n \text{ and } \sum (x_n + y_n)$$

both converge. Since

$$y_n = (x_n + y_n) - x_n,$$

by the algebraic limit theorem, the difference of two convergent series must also converge. That is y_n must converge, so the statement is false.

(d) This is possible, consider

$$x_n = \begin{cases} 0, & \text{if } 2n-1 \\ \frac{1}{n}, & \text{if } 2n \end{cases}$$

Then $0 \leq x_n \leq \frac{1}{n}$ holds. For $\sum(-1)^n x_n$, since $x_n = 0$ for odd n, we have

$$\sum(-1)^n x_n = \sum \frac{1}{2n} = \frac{1}{2}\sum \frac{1}{n}.$$

Since $\sum \frac{1}{n}$ is the harmonic series,

$$\sum(-1)^n x_n \quad \text{diverges.}$$

Exercise 2.7.7

(a) Suppose $a_n > 0$ for every $n \in \mathbb{N}$ and $na_n \to l$ with $l \neq 0$. Since $a_n > 0$ and $l \neq 0$, it follows from the ordered limit theorem that $l > 0$. Let $\epsilon > 0$, since $na_n \to l$ choose $N \in \mathbb{N}$ such that

$$n \geq N \implies |na_n - l| < \frac{l}{2}.$$

That is

$$-\frac{l}{2} < na_n - l < \frac{l}{2}.$$

So

$$na_n > \frac{l}{2},$$

and

$$a_n > \frac{l}{2n}.$$

Then

$$\sum \frac{l}{2n} = \frac{l}{2}\sum \frac{1}{n},$$

which diverges since it is a constant multiple of the divergent harmonic series. Therefore, by the comparison test

$$\sum a_n \quad \text{diverges.}$$

(b) Assume $a_n > 0$ for all $n \in \mathbb{N}$ and $n^2 a_n \to L$ for some $L \in \mathbb{R}$. Since $n^2 a_n$ converges, it is also bounded. So there exists $M > 0$ such that

$$|n^2 a_n| \leq M$$

for all $n \in \mathbb{N}$. Since $a_n > 0$, we also have

$$0 < n^2 a_n \leq M.$$

Dividing through,

$$0 < a_n \leq \frac{M}{n^2}.$$

Then since $\frac{1}{n^2}$ is a convergent p-series, it follows that

$$\sum \frac{M}{n^2} = M \sum \frac{1}{n^2}$$

is also convergent. Therefore, by the comparison test,

$$\sum a_n \quad \text{converges.}$$

Exercise 2.7.10

(a) Notice that the infinite product can be rewritten as

$$\prod_{n=1}^{\infty}(1 + \frac{1}{2^{n-1}}).$$

So we let

$$a_n = \frac{1}{2^{n-1}}.$$

Then

$$\sum a_n = \sum \frac{1}{2^{n-1}}$$

is a convergent geometric series since $|r| < 1$.

(b) Notice that the infinite product can be rewritten as

$$a_n = \prod_{k=1}^{n} \frac{2k-1}{2k}.$$

Since $2k > 2k - 1$ and both $2k$ and $2k - 1$ are positive, it follows that

$$0 < \frac{2k-1}{2k} < 1$$

for all $k \in \mathbb{N}$. So the products are bounded. Then

$$0 < \frac{2k+1}{2k+2} < 1,$$

and

$$a_{k+1} = a_k \cdot \frac{2k+1}{2k+2}.$$

So

$$a_{k+1} < a_k,$$

and the products are decreasing. Since the products are bounded below and decreasing, the Monotonic Convergence Theorem says that the products converges to some $L \geq 0$.

Notice

$$\frac{2k-1}{2k} < \frac{2k}{2k+1}$$

for every $k \in \mathbb{N}$. So

$$a_n \leq \prod_{k=1}^{n} \frac{2k}{2k+1}.$$

Moreover

$$\begin{aligned} a_n^2 &\leq \left(\prod_{k=1}^{n} \frac{2k-1}{2k}\right)\left(\prod_{k=1}^{n} \frac{2k}{2k+1}\right) \\ &= \prod_{k=1}^{n} \frac{2k-1}{2k+1} \\ &= \frac{1}{3} \cdot \frac{3}{5} \cdot \frac{5}{7} \cdots \frac{2n-1}{2n+1} \\ &= \frac{1}{2n+1}. \end{aligned}$$

Thus,

$$0 < a_n \leq \frac{1}{\sqrt{2n+1}}.$$

Since

$$\frac{1}{\sqrt{2n+1}} \to 0,$$

the Squeeze Theorem implies

$$a_n \to 0.$$

(c) Exercise 2.7.10 (c) Proof. Define the partial products

$$a_n = \prod_{k=1}^{n} \frac{4k^2}{4k^2-1}.$$

Since

$$\frac{4k^2}{4k^2-1} > 1,$$

we have

$$a_{n+1} = a_n \frac{4(n+1)^2}{4(n+1)^2-1} > a_n.$$

Thus, (a_n) is increasing. Now observe that

$$\frac{4k^2}{4k^2-1} = 1 + \frac{1}{4k^2-1}.$$

For every $k \geq 1$,

$$0 < \frac{1}{4k^2-1} \leq \frac{1}{3k^2},$$

because $4k^2 - 1 \geq 3k^2$. Since

$$\sum_{k=1}^{\infty} \frac{1}{3k^2}$$

converges, the Comparison Test implies that

$$\sum_{k=1}^{\infty} \frac{1}{4k^2-1}$$

converges. Therefore, by Exercise 2.4.10(b), the sequence of partial products

$$a_n = \prod_{k=1}^{n} \left(1 + \frac{1}{4k^2-1}\right)$$

converges. Hence,

$$\prod_{k=1}^{\infty} \frac{4k^2}{4k^2-1} \text{ converges.}$$

Exercise 2.7.11 Let $N_k = 2^{2^k}$ for $k \geq 0$. For $N_k \leq n < N_{k+1}$, define

$$a_n = \begin{cases} \dfrac{1}{N_{k+1}} + 2^{-n}, & k \text{ even}, \\ \dfrac{1}{N_{k+2}} + 2^{-n}, & k \text{ odd}, \end{cases}$$

and

$$b_n = \begin{cases} \dfrac{1}{N_{k+2}} + 2^{-n}, & k \text{ even}, \\ \dfrac{1}{N_{k+1}} + 2^{-n}, & k \text{ odd}. \end{cases}$$

Set $a_1 = b_1 = 1$. Both sequences are strictly positive and strictly decreasing because their block values are nonincreasing and 2^{-n} is strictly decreasing. For every even k,

$$\begin{aligned} \sum_{n=N_k}^{N_{k+1}-1} a_n &\geq \frac{N_{k+1} - N_k}{N_{k+1}} \\ &= 1 - \frac{N_k}{N_{k+1}} \\ &\geq \frac{1}{2}. \end{aligned}$$

Since there are infinitely many even k, $\sum a_n$ diverges. Similarly, considering odd k, $\sum b_n$ diverges. However, for $N_k \leq n < N_{k+1}$,

$$\min\{a_n, b_n\} = \frac{1}{N_{k+2}} + 2^{-n}.$$

Thus,

$$\begin{aligned} \sum_{n=N_k}^{N_{k+1}-1} \min\{a_n, b_n\} &= \frac{N_{k+1} - N_k}{N_{k+2}} + \sum_{n=N_k}^{N_{k+1}-1} 2^{-n} \\ &\leq \frac{1}{N_{k+1}} + \sum_{n=N_k}^{N_{k+1}-1} 2^{-n}. \end{aligned}$$

Since

$$\sum_{k=0}^{\infty} \frac{1}{N_{k+1}} \leq \sum_{k=0}^{\infty} \frac{1}{2^{k+1}}$$

and $\sum_{n=2}^{\infty} 2^{-n}$ both converge, the Comparison Test implies

$$\sum_{n=1}^{\infty} \min\{a_n, b_n\} \text{ converges.}$$